APS Battery Feasibility Lab

Project selection and full research blueprint

Prepared October 10, 2026 | Arizona / Arizona Public Service

Selected project: a source-backed battery development and dispatch lab that compares 4-, 6-, and 8-hour storage, evaluates peak-hour support, and calculates a hypothetical fixed contract fee required to cover stated costs.

The decision it serves: Which storage duration is worth investigating for an APS-oriented project, what commercial assumptions support it, and what evidence must be obtained before selecting a site or making an investment decision?

This release includes working Python code, 14 tests, measured AZPS summer backtests for 2024-2026, synthetic demonstrations, public-data adapters and a development roadmap. It is an independent research prototype. Measured inputs are revised EIA balancing-authority demand; dispatch responses are simulated. All project costs remain illustrative. No site study or negotiated bid is claimed.

The measured-data extension is documented in RESEARCH_UPDATE.md and the executed notebook. Four-hour prior-week period-peak reductions are 99.5, -0.5 and 34.5 MW in the three summers. Longer duration does not improve that metric in this screen. One EIA-adjusted warmup hour and a reported-value sensitivity are retained. Use the measured extension when presenting current findings; the synthetic chapter below remains a separate demonstration of broad-plateau behavior.

The project is designed around a useful professional question: can you connect the physical performance of a battery to the commercial terms of a development case while preserving the difference between observed data and assumptions? That is the central theme of every module below.

Read in order: project choice; APS context; version-one scope; datasets; battery physics; evaluation; economics; development evidence; implementation; tests; results; roadmap; communication; source register.

1. Why this is the strongest project

The project needs to be relevant to APS, buildable with public information, and substantial enough to demonstrate both technical and commercial judgment. The following comparison is qualitative judgment, not a measured scoring system.

Candidate	Strongest feature	Main limitation	Decision
Battery trading optimizer	Concrete physics and optimization	Arizona market access and revenue assumptions need careful treatment	Keep dispatch as a module
Electricity-price ML model	Forecasting and evaluation skills	APS-specific tradable prices and settlement rights are incomplete	Add later after defining the decision
Arizona grid map	Visible local infrastructure context	Proximity does not establish interconnection capacity	Add as an evidence layer later
Financial model alone	Directly connects costs and contracts	Unsupported revenues can make precise returns misleading	Connect to physical performance
APS storage feasibility and dispatch lab	Links performance, public context and contract costs	Cannot replace private studies and bids	Selected

The combined lab stays focused on one decision: storage-duration screening. Mapping, forecasting and finance are useful only insofar as they improve that decision. A polished map with no capacity evidence, or a large neural network with no operational use case, would be less convincing for this objective.

The initial comparison holds nominal modeled power at 100 MW and varies usable AC-output energy: 400, 600 and 800 MWh. This isolates duration. These are research configurations, not APS-approved designs or a recommended construction scale. In particular, net contractual delivery must account for auxiliaries and losses at the actual delivery meter.

The first result should be a defensible explanation of what changes with duration: peak-hour coverage, recharge requirements, capital cost and sensitivity to forecast error. It should not be an unsupported declaration that the largest battery wins. The next duration increment is useful only when its incremental value justifies its incremental cost and development constraints.

2. APS-specific context and what it changes

APS's published 2025 solicitation sought at least 1,000 MW for operation mainly in 2029-2031. Its March 18, 2026 proposal deadline has passed. The September 30 RPAC update reported 51 shortlisted proposals, including alternative battery durations. These support studying duration now; they do not establish an open bidding opportunity. [S2-S4]

The RFP's storage reference includes four-hour full-capacity discharge under extreme design conditions, with longer duration preferred if cost competitive; summer/non-summer availability requirements are 98%/97%. It includes storage tolling as a structure. These are design references, not a claim this prototype meets every requirement. [S3]

A Q1 2026 filing reports 3,606 MW of storage contracted through the 2023 procurement, expected in service 2026-2028 at that filing. Keep contracted, under-construction and operating resources separate. [S5]

APS has announced expected Markets+ participation as early as 2027. Earlier CAISO material documents APS's imbalance-market implementation. A future market plan does not justify assuming either an ERCOT-style revenue stack or every current CAISO product is available to an independent Arizona asset. [S6-S7]

These facts lead to three modeling choices. First, study peak support and contract-cost screening before merchant-profit forecasting. Second, compare duration and heat-performance assumptions explicitly. Third, date every market and procurement input so that a later rule or planning change can be substituted without silently rewriting the historical analysis.

APS serves a wider area than Phoenix and is not synonymous with every Arizona utility territory. A Phoenix-area parcel must be verified against the actual utility, transmission owner and relevant permitting authority. An Arizona map alone cannot establish that APS is the correct buyer or interconnection provider.

Draft 2026 planning presentations are treated as context, not final adopted resource additions. The lab does not reproduce or rely on APS's internal production-cost model. Its conclusions are bounded by the public evidence.

3. Executable core and current measured-data extension

The executable core has two separate analyses: a physical dispatch experiment and a developer fixed-fee cost screen. They share size assumptions but do not pretend that a demand-shape metric is a commercial payment.

Implemented module	Input	Output
Data validation	Hour-start UTC CSV	Complete, hourly Phoenix-local days
Battery MILP	Daily demand forecast, equipment assumptions	Charge, discharge and energy state
Forecast comparison	Prior-week same-hour load	Actual-day performance without target-day lookahead
Hindsight comparison	Actual daily load	Comparator for the same daily problem
Duration sweep	4, 6 and 8 usable hours	Six matched-period dispatch comparisons
Cost screen	Explicit capex/O&M/augmentation assumptions	Required nominal-kW monthly fee
Sensitivities	Energy cost and upgrade cost	27 scenario rows
Audit outputs	Configuration, input hash, run timestamp	CSV records, figures and manifest

The simulation is a single analytical node. It minimizes a daily load-envelope proxy with full-day foresight only for the labeled hindsight comparator. It does not dispatch APS's generation fleet or model transmission limits. The finance model assumes a hypothetical contract in which the buyer bears charging energy and variable dispatch costs; the owner covers fixed costs and one augmentation. That allocation must be changed if a real contract says otherwise.

Version one does not compute ELCC, utility procurement savings, nodal congestion, merchant revenue, retail tariff savings or equity IRR. It does not contain a manufacturer aging curve, a site recommendation, tax benefits or lender terms. Those are substantive extensions that require evidence, not cosmetic toggles.

The project is deliberately useful offline. The demonstration is reproducible without an API key. A real-data run is a separate action with an explicit source label, and its output folder is distinct. Publishing the repository does not deploy a website or operate a physical battery.

4. Data plan and source contracts

The first historical dataset should be hourly AZPS balancing-authority demand from EIA-930. EIA's public AZPS monitor establishes the available reporting context, but no measured hourly series was downloaded for this release. [S8] Use at least three complete years for a later substantive analysis, including multiple summers. Hold out the latest complete year after developing the model.

Dataset	First use	Important qualification
EIA-930 AZPS demand	Load shapes and peak-hour experiments	Balancing-area load is not automatically retail-only APS load
AZPS reported forecast	Optional later baseline	Confirm issue time and preserve the original vintage
Weather observations	Describe heat events	Observed future weather is not an operational forecast
Archived weather forecasts	Later real forecasting	Include issue timestamp and location
Public resource records	Operating/contracted context	Verify ownership, status and commercial-operation dates
APS planning and procurement	Requirements register	Track publication date and revision
Queue and site evidence	Later development screen	Snapshot and study outcome matter; distance is insufficient
Vendor and financial documents	Later economics	Quote scope, exclusions and date must be explicit

Canonical input fields are timestamp_utc, load_mw and optional availability. Timestamps label the start of a one-hour interval and include a timezone offset. EIA-930 reports hour-ending UTC; the adapter subtracts one hour. The EIA API can label one-hour quantities as MWh: dividing by one hour yields average MW. The original reporting convention and source units must be preserved. [S9-S10]

The adapter filters AZPS and demand type D, pages through the response and rejects an incomplete requested window. The required key remains in the shell environment. It is not written to manifests or source files. Pagination and conversion are mock-tested; the authenticated network path has not been live verified.

Every run should retain the raw snapshot, retrieval date, exact filters, input hash and assumptions. Do not mix a revised historical actual with a forecast whose historical issuance is unknown and call the result an as-of trading test.

5. Data quality and Arizona time

Use UTC as the storage convention and America/Phoenix for this initial daily grouping. The model never labels Arizona data as California time. A later site in a jurisdiction with a different daylight-saving convention needs its own confirmed timezone; do not generalize Phoenix's convention to every location.

Version one's loader rejects duplicate timestamps, missing intervals, null or negative load, timezone-naive strings, non-hourly spacing and incomplete local days. It requires eight days: seven for the prior-week forecast history and at least one for evaluation. It performs no silent interpolation.

Check	Current behavior	Later research upgrade
Duplicates	Reject	Resolve against documented source revision rules
Missing hours	Reject	Recover source records or disclose bounded repairs
Null demand	Reject	Preserve missingness and quantify excluded exposure
Partial days	Reject	Fetch exact local-day boundaries
Extreme demand	Retain if finite/nonnegative	Investigate outliers against source and weather
Revised observations	Record retrieval date	Retain vintage history for operational replay
Availability	Defaults to assumed 1	Supply measured or scenario-specific outages

Do not fill missing hot-afternoon demand with an average day. Missingness can be most damaging in the exact hours used to claim storage usefulness. Report the number of complete summers, each year's coverage, the largest gaps and whether extreme weather dates are included before interpreting performance.

EIA reporting also has storage-accounting conventions: known charging can be excluded from D and reported as a negative contribution to net generation. Therefore load + charge - discharge + auxiliary consumption is this project's analytical load envelope; it is not a prediction of the new published EIA D series or APS's residual generation requirement. [S9]

Do not subtract every EIA solar-generation observation from demand and label it APS net load. Ownership, exports, distributed generation treatment and existing fleet operation require reconciliation. Any future residual-load model should state its equation, resource boundary and source treatment separately.

6. Battery physics and optimization

Power measures the discharge rate; energy measures how much can be delivered. 100 MW at the modeled AC-output boundary for four hours corresponds to 400 MWh of usable AC-output energy. Raw installed DC cell capacity is a different quantity and may need oversizing for usable range, conversion losses, aging and contractual obligations.

The model stores energy in an AC-deliverable coordinate. With one-hour intervals:

E[t+1] = E[t] + RTE * C[t] - D[t]

Here E is usable MWh, C and D are charging/discharging MW at the modeled AC boundary, and RTE is the recharge-to-later-discharge efficiency. This convention places the round-trip loss on charging. Do not separately apply the same loss again to discharge. A later DC-coordinate model should use separate charge and discharge efficiencies consistently instead.

Constraints enforce 0 <= E <= usable capacity, bounded charge/discharge power, exclusive modes and equal starting/ending energy for each local day. A binary mode variable prevents simultaneous charging and discharging. There is no free depletion of initial stored energy at the end of the experiment.

The optimizer minimizes z, where every forecast load-envelope hour is <= z, with a small throughput tie-breaker. This is a peak-shape experiment, not an energy-price objective. SciPy's MILP solver supplies the optimization interface; the code independently checks conservation and physical bounds afterward. [S14]

Auxiliary consumption is an explicit constant grid burden in this version. At 100 MW modeled discharge and 0.5 MW auxiliaries, maximum net support is 99.5 MW. The model does not certify a 100 MW delivery-meter commitment. Uniform power derating and known availability multiply equipment power; they do not replace vendor ambient-temperature curves or unknown outage behavior.

Initial and terminal state are 50% of usable energy every day. This simplifies comparison but prevents multiday scheduling and is not an assertion about APS operations. The first important extension is SOC carryover with a rolling 48-72 hour horizon and a defensible terminal energy value or reserve target.

7. Backtesting and performance measures

Use identical evaluated days for every duration and strategy. The lag-7 schedule uses the same hourly observations from seven days earlier. None of the target day's actual load is used to form that schedule. Actual target-day load is used after scheduling to evaluate the result.

The hindsight comparator uses actual daily load and is labeled accordingly. It answers how well this specific daily problem could perform with demand foresight. It is not a forecast, a global multiyear optimum or an achievable profit estimate. The more limited forecast may select the wrong discharge hours, and a larger battery can still perform poorly under forecast error.

Metric	Definition	What it does not establish
Period peak reduction proxy	Max baseline minus max adjusted over matched days	Accredited capacity or avoided cost
Mean daily reduction proxy	Average daily baseline peak minus adjusted peak	Annual firm capacity
Worst daily reduction proxy	Minimum daily reduction, negatives retained	Extreme-system reliability
Top-100-hour support	Average D-C-aux at highest baseline hours	Utility-defined critical-hour payment
Forecast MAE	Mean absolute actual/lag-7 error	Dispatch value by itself
Equivalent usable cycles	Total discharged MWh / usable MWh	Warranty life under arbitrary definitions
Charge/discharge energy	Hourly totals using one-hour intervals	Market settlement or revenue

Historical actuals downloaded later can be revised. Avoiding target-day lookahead is useful but does not make the test a true historical information-set replay. The current model also knows each day's supplied availability profile in advance. Unknown forced outages and their correlation with heat must be tested separately.

Later baseline improvements should include a fixed schedule, utility reported day-ahead forecast if its issuance is validated, and a weather-based model. Compare out-of-sample peak-hour errors and dispatch outcomes, not just overall MAE. Keep a simple method if a sophisticated forecast fails to improve the actual decision. Never tune the final model on the held-out evaluation year.

System adequacy requires fleet and outage context beyond this experiment. A claim of ELCC or loss-of-load improvement needs an adequacy model and should be reported as a separate result with its own assumptions and validation.

8. Developer economics: the fixed-fee screen

The first commercial model asks what hypothetical level fee would cover a specified cost stream at a chosen unlevered discount rate. It does not claim APS would pay that fee. This inverse calculation is more transparent than inventing a contract price and reporting a precise investment return.

Assume a buyer-controlled dispatch contract, with buyer-paid charging energy and variable dispatch costs. The seller pays initial capital, annual fixed O&M and one augmentation. This is a research contract allocation; verify ownership of energy, losses, auxiliary consumption, market proceeds and performance liability in any real term sheet before changing cash flows.

Capex = (power kW * power cost + usable AC kWh * energy cost + interconnection + development) * (1 + contingency)

Annuity factor = sum of 1 / (1 + real discount rate)^year

Required fee = PV(all modeled costs) / (nominal power kW * 12 * assumed paid availability * annuity factor)

The resulting fee is USD per nominal kW-month in constant real dollars. The code checks that the cash-flow NPV at that fee is approximately zero. The assumed availability multiplier is a simplification, not the negotiated APS settlement formula or evidence that a project satisfies contractual availability.

Input	Illustrative base	Meaning
Nominal modeled power	100 MW	Battery AC output before auxiliaries
Power-related capex	$350/kW	User-adjustable assumption
Energy-related capex	$180/usable AC kWh	User-adjustable assumption
Interconnection allowance	$15 million	No utility study obtained
Development allowance	$5 million	No site budget obtained
Contingency	10%	Applied to the listed initial subtotal
Fixed O&M	$20/kW-year	No vendor operating quote obtained
Augmentation	$30/usable AC kWh in year 10	Stylized payment, not a battery-aging model
Horizon / real discount rate	20 years / 8%	Screening assumptions
Paid availability multiplier	98%	Commercial assumption only

These figures are intentionally inspectable placeholders. A historical ATB cost framework informs how power and energy costs can be separated, but no configuration number is represented as an ATB value or current quote. [S13]

9. Economics that must be added before investment use

Separate the buyer's system value from the developer's cash flow. A buyer may value capacity, reduced production cost, flexibility and reliability. A developer gets the payments allocated by a contract and bears its own costs and liabilities. Those are different accounting boundaries.

Do not add a full merchant-profit forecast to a fixed-fee contract whose buyer owns dispatch and energy proceeds. Add only contractually retained revenue. Do not add retail demand-charge savings to a utility-scale front-of-meter case without a valid tariff and customer-meter boundary.

Missing item	Why it could change the conclusion	Evidence needed
Network upgrades	Can dominate incremental duration economics	Actual studies and cost allocation
Heat/aging guarantees	May require extra installed cells or power	Vendor curves and delivered-energy warranty
Collateral	Creates financing and liquidity requirements	Current procurement/contract terms
Debt and reserves	Changes equity cash flow and constraints	Lender term sheet and covenants
Taxes and credits	Changes eligible cash-flow timing	Current project-specific tax analysis
Land and property costs	Site-dependent annual and upfront burden	Lease/option, assessor and insurance inputs
Energy and loss allocation	Changes dispatch-linked costs	Metering and commercial agreement
Availability penalties	Can make a nominal fee inadequate	Definitions, exceptions and damages schedule
Construction timing	Delays income and adds carrying costs	EPC milestones and funding schedule
Decommissioning	Adds end-of-life obligations	Site/contract requirements and budget

Use real discount rates with constant-real-dollar cash flows, or nominal discount rates with escalated nominal cash flows. Mixing them distorts the required fee. Version one uses constant real dollars and does not estimate interest during construction, cost inflation or salvage value.

Run targeted scenarios before probabilistic simulations. Change interconnection, energy capex, duration, heat-power factor, efficiency, life and augmentation. If a single unsupported input controls the conclusion, report that dependency. A Monte Carlo distribution over invented inputs does not create new evidence.

After real load and verified costs are available, show the incremental 6-versus-4 and 8-versus-6 case: additional cost, additional useful performance, required commercial value, and the missing facts that could reverse the ranking.

10. Site and interconnection evidence

The first site feature should be an evidence register, not a colored map that implies available grid capacity. APS's public FERC interconnection page points to its tariff/OASIS materials; no study or available-capacity conclusion was obtained for this release. [S11]

Start with three hypothetical site categories: an existing generation site, an industrial-area candidate, and a new rural parcel. Do not attach a real parcel name, owner or preferred location until the required evidence is checked. These categories are comparison cases, not development recommendations.

Field	Purpose	Acceptable evidence
Territory and network owner	Identify correct counterparties	Utility map plus specific confirmation
Point of interconnection	Define physical delivery	Identified bus, ownership and documented study path
Capacity and upgrades	Establish technical feasibility	Applicable utility/network studies
Land control	Establish development rights	Signed option/lease and access rights
Zoning and jurisdiction	Identify permitting authority	Current parcel and jurisdiction records
Fire and emergency access	Validate design approval path	Current authority review and site design
Drainage and environmental risks	Identify constraints	Relevant maps, surveys and specialist findings
Equipment and civil access	Assess logistics	Site plan and construction scope
Schedule and cost confidence	Test commercial-operation timing	Study, procurement and permitting milestones

Phoenix publishes battery/solar fire-permit resources, but their availability does not mean a particular utility-scale design is approved or that every Arizona parcel falls under Phoenix rules. [S12] Design-level safety and legal conclusions are outside this code's scope.

Use status values such as unknown, preliminary, verified or blocking. Unknown interconnection cost is a visible unknown, not zero. A short straight-line distance to a transmission line is not proof of a viable connection. Any future ranking should keep unverified candidates unranked rather than burying missing technical facts in a numerical score.

A useful site report explains the next study that could resolve uncertainty and the development budget at risk before resolution. It does not represent itself as an engineering interconnection study or a permit approval.

11. Implementation and how to run it

The repository has a small executable core so each module can be understood. lab.py contains the equipment model, validator, dispatch MILP, forecast baseline, fixed-fee calculation and plotting. fetch_eia.py is a separate optional public data adapter. config.json holds assumptions; test_lab.py checks the important behaviors. README.md is the entry point for a GitHub reviewer.

Create a Python virtual environment, install requirements, run the tests and then run lab.py. The default demonstration needs no external account. It writes synthetic_input.csv, hourly_dispatch.csv.gz, dispatch_summary.csv, finance_screen.csv, cost_sensitivity.csv, a run manifest and two plots into results/demo.

For a later measured-data run, set your own EIA_API_KEY in the environment and run fetch_eia.py with a Phoenix-local inclusive start date and exclusive end date. Then pass the resulting CSV to lab.py with an explicit source-label and a separate results/real directory. The authenticated API path must first be verified on the machine performing the download.

Function	Plain-language role
Battery	Defines power, energy duration and physical assumptions
validate_frame	Enforces the hourly CSV data contract
solve_day	Chooses physically feasible charge/discharge hours
forecast_day	Builds the simple same-hour prior-week forecast
finance	Finds the hypothetical cost-covering fee
run	Executes duration/strategy comparisons and writes evidence
fetch_pages	Retrieves a complete paginated API response
normalize_rows	Converts source labels and units to the canonical contract

The optimizer's constraints are intentionally visible in code. If you explain the project to someone, start with power versus energy, then the state update, then the scheduling objective. Explain the distinction between the physical experiment and the financial screen before showing a monetary number.

No remote repository has been created as part of this preparation. The connected account was jfasoxx; the available integration needs an existing repository URL. Publication should include code, source register, guide and labeled demo output. API keys, virtual environments and later private data must remain excluded.

12. Verification and what a passing test means

The current 14 automated tests passed during preparation. They verify energy conservation, recharge losses, no simultaneous modes, unchanged terminal energy, dispatch shutdown during a known outage, heat-power limits, the flat-load sanity case, duration behavior under hindsight, invalid-input rejection, target-day lookahead exclusion, the financial break-even identity and API adapter behavior.

These tests establish internal model properties. They do not verify an APS contract, vendor product, utility data accuracy or a site. The optional network downloader was tested with simulated API responses, not a live key.

Validation stage	Acceptance criterion
Model physics	No bound, conservation or mode violation in exported records
Temporal evaluation	All strategies use the same post-warmup observations
Finance algebra	NPV at required fee equals zero within numerical tolerance
Data provenance	Exact input hash, source label and timestamp convention saved
Visual QA	Readable chart/PDF labels and explicit data-mode markings
Real-data extension	Source totals and extreme-day coverage reconciled
Forecast extension	Issue-time inputs verified; held-out performance reported
Development extension	Evidence distinguishes unknowns from verified constraints

Before accepting a real-data conclusion, compare source rows with the monitor on selected normal and extreme days. Check timezone shifts manually. Confirm that the evaluation interval includes charging as well as discharge and that initial/final energy treatment does not give one strategy an advantage.

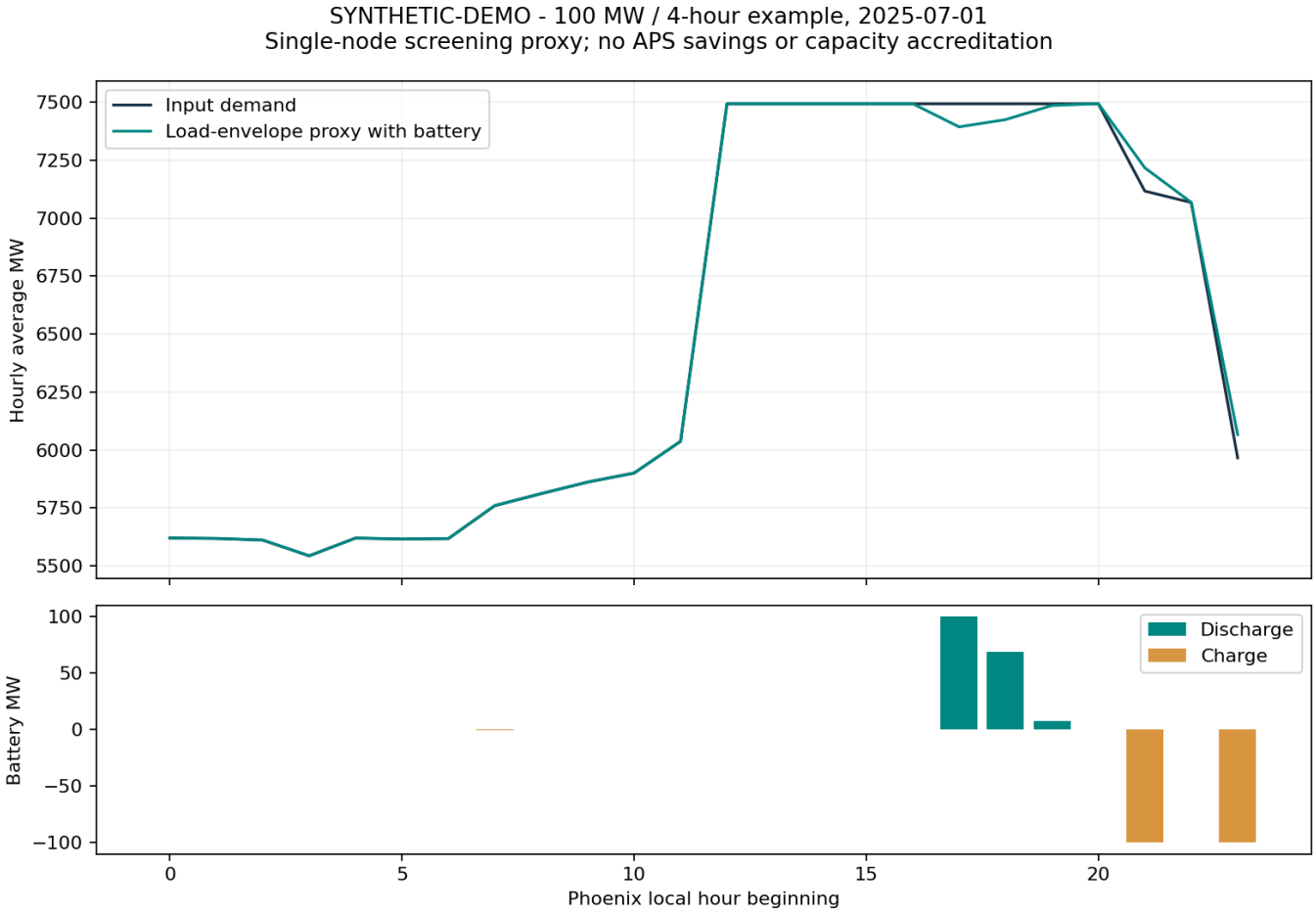
Audit commercial units: MW versus kW, MWh versus kWh, USD/kW-month versus USD/MWh, usable capacity versus raw installed capacity. A thousand-fold unit error can survive a visually convincing presentation.

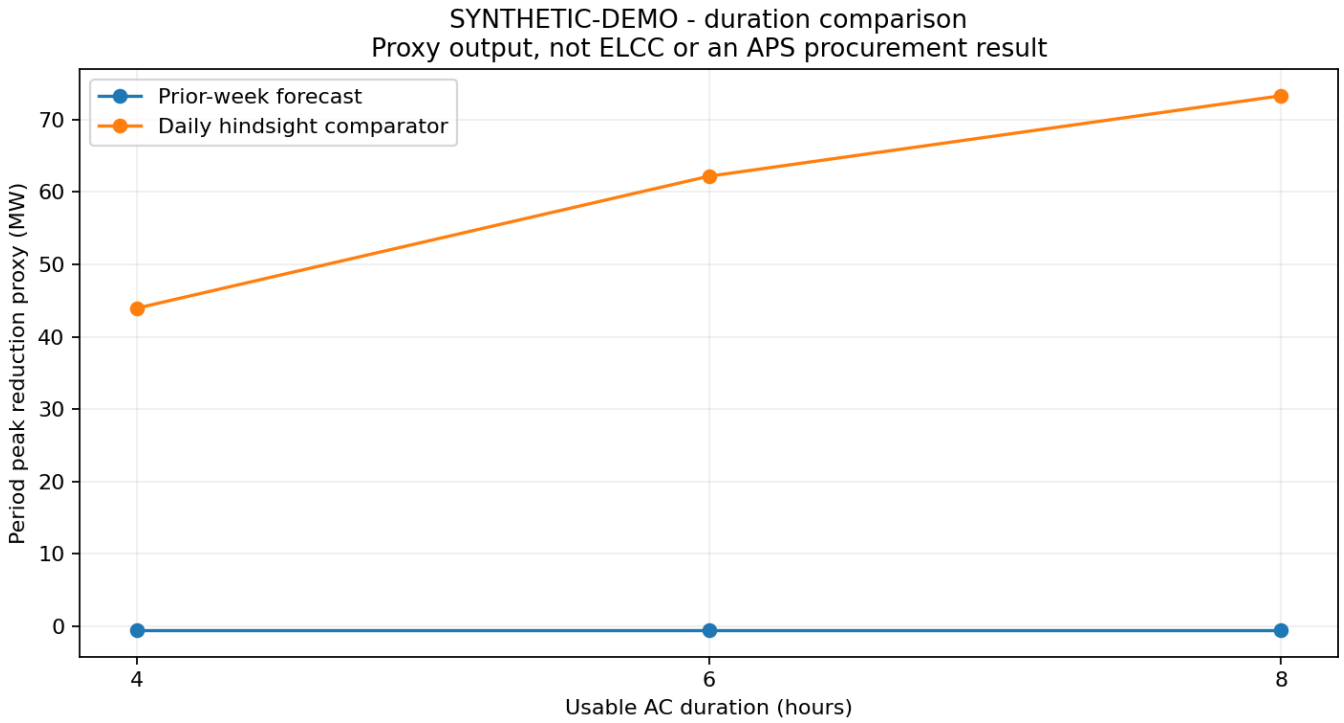
When extending the model, test the new decision-relevant behavior rather than just its implementation. For rolling dispatch, test conservation across day boundaries. For aging, test that guarantees can be sustained or fail visibly. For a contract penalty, test the cash-flow impact of the defined event. Keep the old tests so added features do not quietly weaken the original guarantees.

13. How to interpret the included demonstration

The demonstration fabricates 42 summer-style days, including narrow peaks and broad evening plateaus. Seven days provide forecast history and 35 are evaluated. The shapes are purpose-built to exercise the model. They are not calibrated to APS history and should not be used to recommend an Arizona project size.

The two figures below are synthetic outputs from the current implementation. They illustrate schedule behavior and duration comparisons. The actual values can be inspected in the accompanying CSV files; no market revenue is computed.





The forecast result can lag the hindsight comparator because the prior week's load does not match a new broad heat-event shape. That demonstrates the value of evaluating schedules on actual held-out observations, but it is not evidence that an APS forecasting method performs similarly.

If longer duration improves a proxy metric, it still needs an economic and reliability justification. If it does not improve that metric, that alone does not prove it has no value: reserve obligations, duration accreditation and commercial duties are absent from this prototype. Keep each conclusion inside the boundaries of the experiment that produced it.

14. Roadmap to a serious APS research project

The milestones below are suggested work packages, not claims that unfinished modules already exist. Complete each evidence gate before moving to stronger commercial conclusions.

Phase	Work	Completion evidence
A: usable starter	Code, tests, synthetic outputs, primary sources	This release
B: measured history	Three AZPS summer snapshots, correction sensitivity; weather remains open	Summer demand complete with provenance and coverage review
C: operational replay	Forecast vintages, rolling SOC and outages	Matched out-of-sample dispatch results
D: physical realism	Ambient derating, aging, auxiliaries and reserve limits	Vendor assumptions and sensitivity results
E: commercial screen	Contract allocation and verified cost ranges	Clear fee/cost bounds and missing-term list
F: development screen	Territory, site control, study path, permitting	Evidence-backed candidates and unresolved blockers
G: professional review	Technical and commercial practitioner feedback	Recorded fixes and bounded conclusions

A reasonable first working sequence is: run the demo; explain every variable; fetch one complete real year; inspect ten peak days; compare durations with the simple forecast; stress temperature and costs; document the limitations; then add better forecasting or an evidence-based site view.

The measured summer extension is now complete. The next work is full-year and future net-load validation, rolling SOC and heat/outage realism rather than adding decorative maps. Build a web interface only when it makes the actual decisions and source boundaries clearer.

The project can eventually support separate professional questions: resource planning asks about system value; an independent developer asks about the contract and execution path; a lender asks about downside cash flow and guarantees. Keep those analyses separate and tie them to the same documented equipment and source assumptions. Shared inputs do not mean shared conclusions.

The initial deliverable is strong because it is runnable and inspectable. A serious later result will be stronger because verified evidence changes the decision, not because the project claims to do more than its data support.

15. Presenting the project accurately

An accurate GitHub introduction is: "Independent APS-oriented battery research lab comparing storage duration, forecast dispatch and the fixed fee needed to cover explicit illustrative costs. Includes physically constrained optimization, data validation, reproducible synthetic examples and a real-data adapter."

Avoid descriptions such as "built APS's trading system," "found the best Arizona battery site," "proved a profitable 100 MW project," or "reduced APS costs by the chart's MW value." Those would attribute access, validation or commercial results the repository does not have.

When showing the project, be ready to answer these questions:

- What exactly does the load-envelope proxy measure?
- Which data are measured, and which are synthetic?
- Why does longer duration sometimes help and sometimes not?
- How are energy losses and auxiliaries counted?
- How do you prevent future information from entering a forecast?
- Who pays for charging and owns the energy proceeds in the financial case?
- Which cost inputs could reverse the ranking?
- What would an interconnection study establish that a map cannot?
- What is implemented, and what remains on the roadmap?

The current answer to project selection is definite: this combined storage feasibility and dispatch lab is the best of the discussed options for a focused APS-oriented project. The eventual answer to investment selection remains open: actual project economics and feasibility require measured data, quotes, studies and commercial terms that are not yet present.

Consult SOURCES.md for the complete primary-source register, publication dates, source limitations and URLs. All source claims are bounded to their reporting period. No APS source PDF, private attachment, brand asset or confidential input has been redistributed in the repository.

Primary sources and evidence boundaries

S1 - APS resource planning

Public planning materials and the ongoing 2026 IRP process; use dated materials rather than treating the 2023 plan as current.

S2 - APS RFP overview

Published 2025 solicitation sought at least 1,000 MW; target operation 2029-2031, with earlier/later consideration. This is not an assertion that bids remain open.

S3 - 2025 APS All-Source RFP, revised January 6, 2026

Printed pp. 4, 12, 37-39: deadline, storage contracts and technical reference. Proprietary platform appendices were not accessed.

S4 - APS RPAC presentation, September 30, 2026

Slide 7: 51 shortlisted proposals and multiple battery durations. Draft planning scenarios are preliminary; not treated as final approvals or a bid recommendation.

S5 - Pinnacle West / APS Q1 2026 filing

3,606 MW of battery storage contracted through the 2023 procurement, expected in service 2026-2028 at filing. Contracted is not the same as operating; later filings can change status.

S6 - APS Markets+ announcement

Participation expected as early as 2027. A plan is not confirmation of live participation. Future market rules require a separately dated adapter.

S7 - CAISO APS EIM implementation agreement library

Historical APS energy imbalance market implementation documentation; does not establish every present-day settlement or future rule.

S8 - EIA AZPS grid monitor

Public AZPS demand, forecast, generation and interchange overview. Measured snapshots in this release come from the public workbook listed in S17.

S9 - EIA-930 instructions

pp. 3 and 5: hour-ending UTC reporting and treatment of known energy-storage charging. This is physical grid reporting, not a retail tariff or node price series.

Primary sources (continued)

S10 - EIA API technical documentation

API keys, metadata, facets and pagination. The included downloader is mock-tested, not live authenticated end-to-end verified.

S11 - APS FERC generator interconnections

Public entry point for APS OATT/OASIS and interconnection materials. No queue snapshot, system study or available capacity was inferred.

S12 - Phoenix fire permit forms

Municipal battery/solar permit resources. Site-specific jurisdiction and current requirements must be confirmed; Phoenix rules do not govern every Arizona parcel.

S13 - NREL/NLR 2024 ATB utility-scale battery framework

Search-indexed historical power/energy cost and duration framework only. Full page did not open in this session. No numeric costs in this repository are represented as ATB values or current quotes.

S14 - SciPy MILP reference

Solver interface, integrality, bounds and linear constraints. The implementation separately checks physical feasibility.

S15 - 2025 ATB utility-scale battery storage

Reviewed full page October 10, 2026. 85% round-trip efficiency, 15-year lifetime and fixed O&M at 4% of capex including augmentation inform a separate sensitivity. Capital cost numbers remain illustrative and are not extracted ATB values.

S16 - Berkeley Lab Queued Up 2026

June 2026 report with end-2025 queue data. Provides national interconnection-risk context; national outcomes are not APS-specific completion probabilities or upgrade estimates. No queue dataset is redistributed.

S17 - EIA public AZPS workbook

Downloaded and inspected including Published Hourly Data and Notes. Canonical summer extracts use Adjusted demand, retain one explicit 2026 EIA adjustment, and include a raw reported-demand sensitivity. Source hashes and timestamp conversions are recorded beside each CSV.

S18 - GitHub Python validation documentation

Official workflow reference for the repository's reproducibility checks. Local scientific tests and audits are distinct from remote workflow status.

Source claims describe their reporting periods. Project costs are illustrative. Bundled demand includes measured EIA snapshots and a separately labeled synthetic demo. Private APS bids, system studies, proprietary appendices, vendor quotes and negotiated terms are absent.